1 Context, positioning, and aims of the proposition

1.1 Objectives and positioning

The last decade has seen the emergence of diffusion models in machine learning and its rise to a major cultural impact: nowadays, the general audiance is now very familiar with "AI-generated images" through the famous DALL-E, Midjourney or Stable Diffusion applications. Once trained, the diffusion models behind those famous applications construct an image answering a text prompt by starting from some noise and then denoising it incrementally, taking the prompt into account.

A similar idea exists in control theory, known as the backstepping method: construct a feedback controller by recursively transforming a system with known control toward the target system while simultaneously deforming the control. This same deformation idea has been applied to construct polynomial root-finding algorithm in numerical algebraic geometry: it is known as homotopy continuation and has become a standard technique to find approximations of solutions of polynomial systems.

Diffusion models in machine learning, homotopy continuation in numerical algebraic geometry, backstepping in control theory or deformation in fundamental mathematics are all, in some sense, faces of the same polyhedron: building a solution to a target problem by incrementally deforming a solution of a starting system.

In this project, we intend to build a bridge from the ideas of machine learning toward that of homotopy continuation. There is an early popular text on the subject of homotopy continuation and its applications by Morgan [**Mor09**]. More precisely, we consider the problem of solving a square system of n polynomials in n variables having the form

$$\mathbf{P}(\mathbf{x}) = 0, \tag{$\mathcal{P}$}$$

with $\mathbf{P} = (P_1, \cdots, P_n)$ and $\mathbf{x} = (x_1, \cdots, x_n) \in \mathbb{C}^n$. This problem appears widely in scientific computations as optimization problem or cryptosystems. Homotopy continuation methods consist on tracking solutions of equations of $\mathcal{P}$ as they are deformed continuously from a known problem to an unknown problem. The path that connects both problems is called a **homotopy**. Algorithms based on homotopy continuation methods are very efficient to solve ($\mathcal{P}$) numerically when all roots are isolated, see for instance the package HomotopyContinuation.jl in Julia.

Overview of the project: In this project, we will combine reinforcement learning methods with homotopy continuation methods to solve polynomial system.

Task 1 Proof of Concept: find efficient homotopy paths to find all the roots of **univariate polynomials** over $\mathbb{C}$.

Task 2 Find efficient homotopy paths to compute all the complex roots of zero-dimensional **square multivariate polynomial systems** (n variables, n-polynomials, finite number of roots).

For those tasks, a neural network will be used as an environment in a reinforcement learning procedure to learn the "best" homotopy path (in the sense of time of the computation and of the quality of the answer) constructed from a given starting system to a given target system. We propose a roadmap of systems with increasing difficulty: univariate, Pham systems, general systems.

This project will be carried out by a synergy of combined expertise in **machine learning algorithms**, **computer algebra** and **optimization**.

Related works: A first connection between homotopy continuation and machine learning has been established by Petr Hruby, Timothy Duff, Anton Leykin, and Tomas Pajdla in [**HDLP22**]. The authors have used Machine Learning and classifiers to exclude the tracking of a part of the roots in Homotopy Continuation so as only a non-spurious root is obtained. The goal was to save time in a context where only one root can be enough. They applied their method to computer vision

problems and won the best paper award at IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), the premier conference in computer vision. Our approach differs in that we wish to use Reinforcement Learning to design an optimal homotopy path to get all the roots while their work found ways to exclude the tracking of roots that are not relevant to their context.

In his 2025-2026 internship in Limoges, Shun Arakawa has established promising preliminary results on Task 1. See **TODO:** Reference to Arakawa-kun's work in this document.

Positioning of the project with respect to the PHC Sakura 2027: Our project is naturally related to the offer of PHC Sakura 2027; the goal of this project is to create/develop a long-term partnership between Limoges University and Chiba University around polynomial system solving using Machine Learning techniques akin to diffusion models. This project will combine the expertise in mathematical analysis, homotopy continuation, computer algebra and machine learning of researchers from both universities. It will contribute to the training and elevation of a very promising Master student. This project will be the cornerstone for establishing future strong connections between the two universities, including PhD students and internships exchanges.

1.2 Methodology

1.2.1 Homotopy continuation

TODO: Improve this part

Homotopy continuation is a technique to approximate zero-dimensional solution sets of square polynomial systems $F : \mathbb{C}^n \to \mathbb{C}^n$. The idea is that one first forms another polynomial system G related to F in a prescribed way, which has known or easily computable solutions. Then the systems G and F can be connected by setting up a homotopy $H : \mathbb{C}^n \times [0,1] \to \mathbb{C}^n$. An example for this would be the linear homotopy $H(x,t) = (1-t)F + tG$. For a properly formed homotopy, there are continuous solution paths leading from the solutions of G to those of F which may be followed using predictor-corrector methods. There are several software packages publicly available to make computations with homotopy continuation such as Bertini [**BHS13**], PHCpack [**Ver99**] and [**Bre18**].

For the sake of clarity, we provide a simple example with $n = 1$ from [**Wam05**] to illustrate the main concept of polynomial root-finding algorithms based on homotopy continuation. Our goal is to find the roots of $F := P_0$ with $P_0(z) = z^2 - 5$ for any $z \in \mathbb{C}$ using a homotopy continuation method from known roots of the polynomial $G := P_0$ with $P_0(z) = z^2 - 1$ for any $z \in \mathbb{C}$. We consider the following linear homotopy $H : \mathbb{C} \times [0,1] \to \mathbb{C}$

$$H(z,t) = tP_1(z) + (1-t)P_0(z).$$

In this case, we have an explicit formulae of the solution paths. For any $t \in [0,1]$, we have two solutions of $H(z,t) = 0$, namely $z_1^*(t) = \sqrt{5-4t}$ and $z_2^*(t) = -\sqrt{5-4t}$. As t goes from 1 to 0, the roots go from ± 1 to $\pm\sqrt{5}$, the roots of the equation $z^2 - 5 = 0$. An homotopy continuation method consists of numerically tracking the solutions of $H(z,t) = 0$ as t goes from 1 to 0. Let $i \in \{1,2\}$, and Δt an adaptative time step-size.

If $z_i(t)$ is an approximate value of $z_i^*(t)$ one of the roots of $H(z,t)$, one can use $z_i(t)$ as a so-called *predictor* for a root $z_i^*(t-\Delta t)$ of $H(z, t-\Delta t)$. The Newton method can be then applied to compute a so-called *corrector* $z_i(t+\Delta t)$, a better approximation of $z_i^*(t-\Delta t)$.

This step of prediction-correction is then applied successively along the homotopy path starting from $z_i(1) = \pm 1$ so as to obtain in the end approximations $z_1(0), z_2(0)$ of $z_1^*(0), z_2^*(0)$.

One step of this root tracking iteration can be written with the following algorithm.

A simple version of the homotopy algorithm for univariate polynomials can then be written as follows with algorithm 2

This idea can be generalized to square polynomial systems as long as all the polynomial systems on the homotopy $H(x_1, \ldots, x_n, t)$ are square polynomial systems with the same number of roots, all

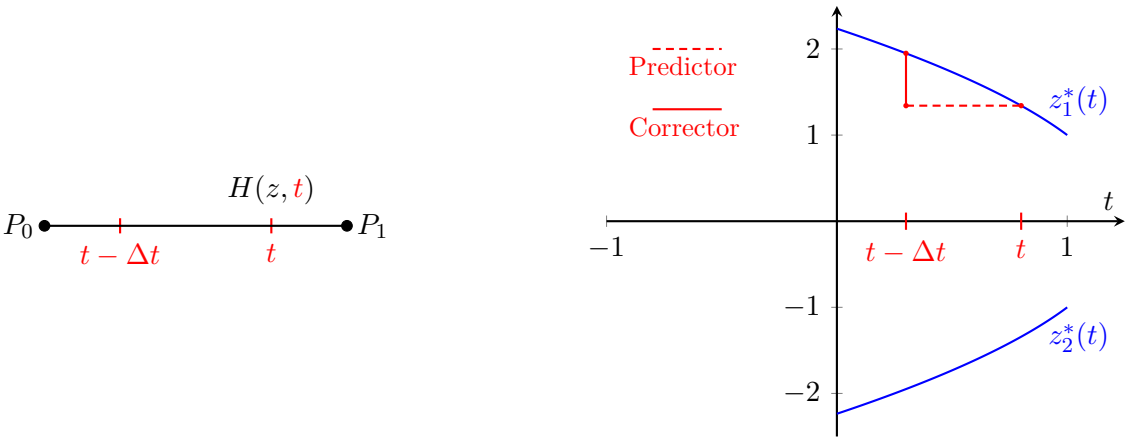


Figure 2. Illustration of one-step polynomial root-finding algorithm based on homotopy continuation method with $P_0(z) = z^2 - 5$ and $P_1(z) = z^2 - 5$. Let $t \in [0,1]$, $z_1^*(t) = \sqrt{5-4t}$ and $z_2^*(t) = -\sqrt{5-4t}$ stands for the two solutions of $H(z,t) = 0$. The value $z_i(t+\Delta t)$ is numerically constructed using a predictor-corrector method. Newton's method may need many iterations.

Algorithm 1 One-step Root Tracking

Let a time $t \in [0,1]$ and an adaptative time step $\Delta t > 0$. Let $H(z,t)$ be a polynomial of degree n in z, with simple roots. We assume we have approximate values $z_i(t)$ of the n roots $z_i^*(t)$ of $H(z,t)$
for $i = 1$ to n **do**
  $z_i(t-\Delta t) \leftarrow$ Newton's method$(H(\cdot, t-\Delta t); z_i(t))$
end for
return $(z_1(t-\Delta t), \ldots, z_n(t-\Delta t))$

simple.

Further generalizations are also possible but are out of the scope of this proposition.

1.2.2 Efficiency and limitations of the homotopy continuation

From the previous presentation, one can infer that the main points governing the efficiency of the method are the following:

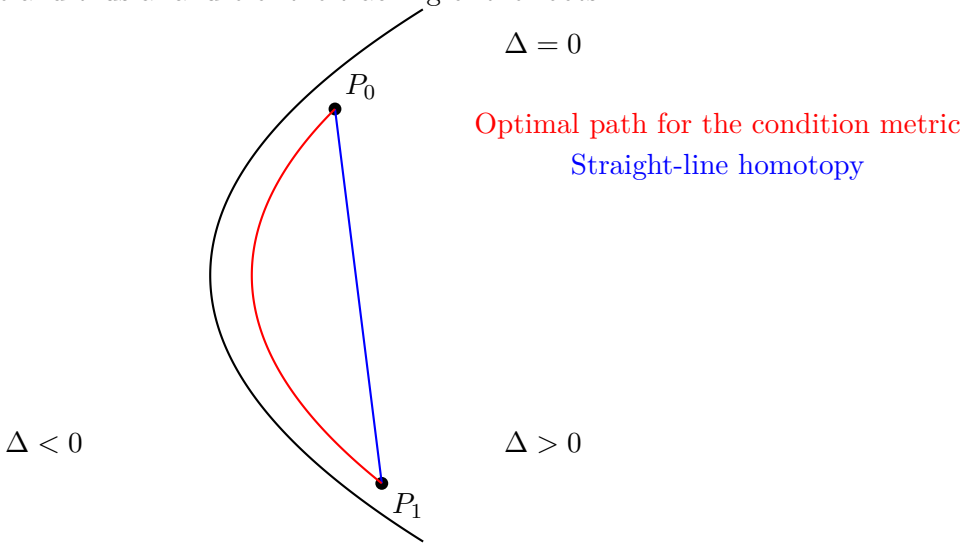
- The choice of the starting point $H(x_1, \ldots, x_n, 1)$.
- The choice of the homotopy $H(x_1, \ldots, x_n, t)$ joining starting point $H(x_1, \ldots, x_n, 1)$ and target point $H(x_1, \ldots, x_n, 0)$.
- The conditionning of the systems $H(x_1, \ldots, x_n, t_i)$:

Algorithm 2 Find roots of a given univariate polynomial P

Let us denote n the degree of P_0.
Choose P_1 with known roots $(r_i^1)_{i\in\{1,\cdots,n\}}$, the step-time Δt and a homotopy H.
Set up a grid $t_0, \cdots, t_M$ with M large enough such that $t_0 = 0$, $t_M = 1$ and for any $j \in \{1, \cdots, M\}$, $\|H(\cdot, t_{j-1}) - H(\cdot, t_j)\| = \Delta t$ and $H(\cdot, t_{j-1})$ admits n roots.
for $j = M$ to 1 **do**
  $(r_i^{j-1})_{i\in\{1,\cdots,n\}} \leftarrow$ Algorithm 1 $\left((r_i^j)_{i\in\{1,\cdots,n\}}, H(\cdot, t_{j-1})\right)$
end for
return $(r_i^0)_{i\in\{1,\cdots,n\}}$

- If $H(x_1, \ldots, x_n, t_i)$ has good conditioning, the *correction* through the Newton method is fast. Larger time-steps Δt may also be used.
- If $H(x_1, \ldots, x_n, t_i)$ has bad conditioning, the *correction* through the Newton method is slow or may fail. Smaller time-steps Δt may also be necessary, meaning more computations.

For monic polynomials of degree 2, written $x^2 + bx + c$, it is well-known that if their discriminant $\Delta = b^2 - 4c$ is small, then they have bad conditioning. Conversely, a big discriminant (in absolute value) means good conditioning. Crossing the discriminant locus means a collapse of the roots into one and thus a failure of the tracking of the roots.



An optimal path for homotopy continuation would then have to optimize the conditioning along the path. It can be formalized in terms of Riemannian metric using the *condition metric*. The geodesics for the condition metric in the parameter space of polynomials will then provide *optimal paths* in terms of efficiency for homotopy continuation. **TODO:** citation

Unfortunately, in practice, the condition metric and its geodesics are **impossible to embrace** in full generality when the degree is bigger than 2.

1.2.3 Task 1: Solve the roots of a polynomial in 1 variable using reinforcement learning along "optimal" homotopy paths

TODO: Complete with Arakawa-kun's internship results and what remains to be done

TODO: adapt the following As already mentioned, the stability of polynomial root-finding algorithms using homotopy continuation in literature and algorithm 2 is highly dependent on the choice of the homotopy H.

Following Arakawa's internship at Limoges in 2025-2026, we have considered using reinforcement learning and the PPO algorithm to train a model that produce near optimal homotopy curves when given starting and target polynomials.

In more concrete terms, the model:

- outputs Beziers control points to define a curve joining P_1 to P_0.
- uses the number of predictor-corrector steps along the path, compared to the case of the straight-line, as a reward.

Here are some preliminary results.

TODO: Add Table from Arakawa's internship

TODO: Add comments

TODO: What about integration to HomotopyContinuation.jl or something like that?

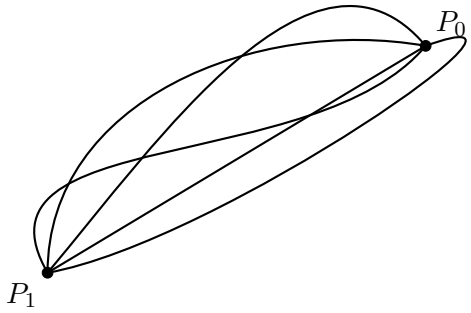


Figure 5. Many trajectories in a polynomial space

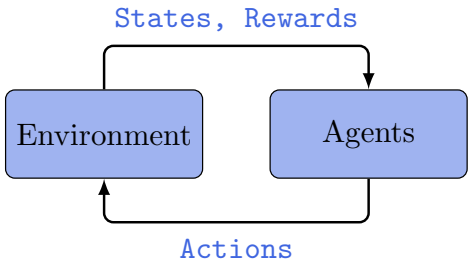


Figure 6. Reinforcement learning procedure

Figure 7. Illustration of a Reinforcement learning polynomial root-finding algorithm

1.2.4 Task 2 : Solve square polynomial systems in n variables using reinforcement learning along "optimal" homotopy paths.

The following goal is to process some more complex systems in a context where one can maintain the following constraints:

- **Zero-dimensional systems**: Only systems with a finite number of solutions are considered.
- **Common monomial basis**: The quotient rings of the start and target systems must share a common monomial basis.
- **Numerical stability**: The condition number must remain manageable throughout the homotopy path and the homotopy path must avoid the singular locus.

To do so, we plan to tackle first the easier case of Pham systems.

*T*2.1**: Pham systems**

Example 1.1 (Pham system)**.** A Pham system is a multivariate polynomial system of the form:

$$f_1 = x_1^{d_1} + h_1, \ldots, f_n = x_n^{d_n} + h_n$$

where $d_1, \ldots, d_n \in \mathbb{Z}_{>0}$ and $h_1, \ldots, h_n \in K[x_1, \ldots, x_n]$ such that for any i, the total degree of h_i is $< \min(d_1, \ldots, d_n)$. Such a system is always of dimension zero.

For simplicity, we begin with a square system in two variables.
Consider a **Fake Bivariate Pham system** of the following form:

$$FBP(x_1, x_2) = \begin{pmatrix} f_1(x_1) \\ f_2(x_2) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \tag{1}$$

where $f_1(x_1)$ and $f_2(x_2)$ are **univariate polynomials**.

Let $\mathcal{Z}(f_1) = \{\zeta_1, \ldots, \zeta_{d_1}\}$ be the set of roots of f_1 (assumed simple), and $\mathcal{Z}(f_2) = \{\xi_1, \ldots, \xi_{d_2}\}$ be the set of roots of f_2 (also assumed simple).

Remark 1.2. The solutions of system (1) are:

$$\left\{ \begin{pmatrix} \zeta_i \\ \xi_j \end{pmatrix} : i \in [\![1, d_1]\!], j \in [\![1, d_2]\!] \right\}$$

This yields $d_1 \cdot d_2$ roots, reaching the **Bézout bound**.

Consider now a more general **bivariate Pham system**:

$$GBP(x_1, x_2) = \begin{pmatrix} x_1^{d_1} + f_1(x_1, x_2) \\ x_2^{d_2} + f_2(x_1, x_2) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \tag{2}$$

where:

$$\max\{\deg(f_1), \deg(f_2)\} < \min\{d_1, d_2\}$$

Task 2.1 Preliminary: find a near optimal path from FBP to GBP.

Task 2.1 General case: find a near optimal path from Fake n-variate Pham system to a General n-variate Pham system

As for Task 1, we plan on training a model for this task using Bézeris curves and the PPO algorithm. One should be aware that the space of parameters for the polynomial homotopy is much larger than in Task 1. Nevertheless the rigid constraints on the definition of Pham systems make it reasonably amenable.

*T*2.2**: General systems** Once Task 2.2 is completed, we will tackle the cases of general zero-dimensional square polynomial systems.

TODO: Consider shape position systems and Kera et al 2024 to put into this shape position

*T*2.3**: Implementation** **TODO:** Good enough implementation to be added to HomotopyContinuation.jl and friends

2 Project organisation and resources allocation

2.1 Team of the project

Composition of the team. The team working on the project consists of the following members (click on names for personal websites):

Team (collaborators)	Current position	Tasks	Contributions
Shun Arakawa	Master student at Chiba University	All tasks	Machine learning (Design and Implementation)
Hiroshi Kera	Associate professor at Chiba University	All tasks	Machine learning coordination
Thibault Liard	MCF at University of Limoges	All tasks	Machine learning (mathematical analysis) and optimisation
Olivier Ruatta	Professor at University of Limoges	All tasks	Homotopy continuation
Tristan Vaccon	MCF at University of Limoges	All tasks	Computer algebra and algorithmics